

NEW RIVER MCAS, NC, USA

WMO: 723096

Lat: 34.708N

Lon: 77.440W

Elev: 8

StdP: 101.23

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93727

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-3.0	-14.6	1.1	-0.5	-12.1	1.3	1.0	11.0	7.9	9.8	8.1	2.7	10	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	33.7	25.4	32.6	25.2	31.5	24.7	26.9	31.1	26.3	30.5	25.7	29.7	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.3	28.9	25.1	20.3	28.2	24.6	19.6	27.8	84.2	31.4	81.4	30.5	79.0	29.9	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.2	7.3	-8.6	36.1	2.1	1.2	-10.1	36.9	-11.3	37.6	-12.5	38.3	-14.0	39.1		
(5)			-9.7	28.5	2.2	0.9	-11.3	29.2	-12.5	29.7	-13.8	30.2	-15.4	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	7.3	9.1	12.4	17.2	21.4	25.3	26.9	26.2	23.6	18.2	12.6	9.4		
	DBStd	8.04	5.87	5.31	5.01	4.27	3.46	2.57	1.88	1.95	2.72	4.23	4.71	5.36		
	HDD10.0	340	123	75	34	2	0	0	0	0	0	1	26	78		
	HDD18.3	1399	343	262	194	71	13	1	0	0	2	55	178	281		
	CDD10.0	3083	41	49	107	219	353	458	522	503	408	257	106	60		
	CDD18.3	1098	1	3	8	37	108	209	264	244	160	52	8	3		
	CDH23.3	9075	3	17	58	252	788	1799	2513	2166	1108	327	35	9		
	CDH26.7	3058	0	2	7	47	218	664	980	778	306	54	1	0		
(14) Wind	WSAvg	3.3	3.5	3.6	3.9	4.0	3.4	3.0	2.8	2.7	3.1	3.0	3.1	3.2		
(15) Precipitation	PrecAvg	1368	104	103	101	80	106	133	187	160	137	74	77	92		
	PrecMax	1780	211	271	203	225	291	300	355	325	568	226	181	167		
	PrecMin	970	36	33	19	13	16	55	88	14	22	16	15	11		
	PrecStd	184	51	59	46	49	64	55	66	83	108	49	43	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.5	25.8	27.6	30.2	33.0	34.9	35.2	35.0	32.4	30.1	26.3	24.6		
		MCWB	18.1	18.8	18.3	19.8	23.5	25.1	26.1	26.1	24.2	22.9	20.2	19.4		
	2%	DB	21.2	22.4	24.9	28.0	30.9	33.2	33.8	32.9	31.0	28.0	24.2	22.4		
		MCWB	17.2	16.7	17.3	19.5	22.9	24.9	25.8	25.7	23.8	22.1	18.8	18.6		
	5%	DB	19.0	20.1	22.8	26.3	28.9	31.9	32.6	31.8	29.8	26.7	22.5	20.5		
		MCWB	16.0	15.8	16.2	18.8	22.1	24.5	25.6	25.4	23.4	21.1	18.3	17.8		
	10%	DB	17.1	18.0	20.9	24.4	27.5	30.4	31.3	30.8	28.5	25.1	20.9	18.7		
		MCWB	14.3	14.2	15.5	17.9	21.2	23.9	25.1	24.8	23.0	20.7	17.6	16.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	20.0	20.3	23.1	25.7	27.0	27.9	28.0	26.1	25.2	22.3	21.2		
		MCDB	21.4	23.5	24.1	26.6	29.9	31.8	32.7	31.8	29.4	27.1	24.6	23.1		
	2%	WB	18.3	18.7	19.1	21.5	24.2	26.1	26.9	26.9	25.4	24.0	20.9	19.5		
		MCDB	20.3	21.3	22.5	25.1	28.4	31.1	31.4	30.7	28.6	26.6	22.6	21.3		
	5%	WB	16.7	16.9	18.0	20.4	23.3	25.4	26.3	26.2	24.8	22.9	19.4	18.2		
		MCDB	18.7	19.1	21.1	24.1	27.1	30.0	30.8	30.0	27.9	25.4	21.5	20.0		
	10%	WB	14.7	14.9	16.8	19.3	22.5	24.7	25.7	25.6	24.2	21.6	18.0	16.5		
		MCDB	16.8	17.7	19.9	23.0	26.0	28.8	29.9	29.2	27.0	24.1	20.2	18.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	11.6	12.0	11.7	10.5	9.5	8.9	8.8	8.9	10.9	11.9	11.0		
		MCDBR	13.1	13.6	13.7	12.8	11.5	10.9	10.1	9.9	10.0	11.2	12.9	12.0		
		MCWBR	9.6	8.8	7.6	6.4	5.2	4.5	4.2	4.3	4.4	5.7	8.2	8.6		
		MCDBR	11.5	12.4	11.2	10.4	9.9	9.8	9.2	8.8	8.1	8.6	10.7	11.1		
	5% WB	MCWBR	9.8	9.5	7.6	6.2	5.0	4.5	4.2	4.1	4.0	5.1	8.3	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.322	0.332	0.362	0.399	0.451	0.493	0.525	0.505	0.423	0.379	0.344	0.328		
	taud		2.524	2.491	2.429	2.318	2.218	2.148	2.073	2.124	2.354	2.452	2.504	2.539		
	Ebn at Noon		887	915	911	887	841	803	774	783	837	850	852	857		
	Edh at Noon		82	94	109	128	143	153	164	153	116	97	83	77		
(44) All-Sky Solar Radiation	RadAvg		2.67	3.39	4.47	5.61	6.15	6.38	6.04	5.48	4.56	3.87	2.95	2.36		
	RadStd		0.17	0.32	0.34	0.35	0.42	0.35	0.35	0.33	0.31	0.43	0.21	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+156	+92
(47) Regional (40 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+159	+115

Nomenclature: See separate page